



CSE111 Course Outline Spring 23 - Salman Sayeed Khan

Computer Programing (Java-1) (BRAC University)



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**Course Outline of
CSE 111: Programming Language II
Semester: Spring 2023**

Course Description:

This course would be an introduction to data structures, formal syntax specification, language theory elements, and mathematical preliminaries. Other topics that would be covered are formal languages, structured programming concepts, and a survey of features of existing high-level languages. Students would design and write applications using an appropriate language. The course includes a compulsory 3-hour laboratory work each week.

Prerequisites: CSE110: Programming Language I

Course Outlines:

Sr. No.	Topic details	Time allocation
1	String, List, Tuple, Dictionary	Week 1
2	Function	Week 2
3	OOP basics: class, object, instance variable & constructor	Week 3
4	Introduction to methods (instance), Method / Constructor Overloading	Week 4
5	Midterm exam	Week 5
6	Operator overloading, Encapsulation	Week 6
7	Class variable and method, Static Method	Week 7
8	Basic concept of Inheritance	Week 8
9	Method overriding, Variable overriding	Week 9
10	Abstraction	Week 10
11	Exception handling	Week 11
12	Review	Week 12

Course Assessment Methods:

Homework

Homework/ assignment shall be designed to ensure that the students have the required knowledge to recognize, analyze, design and apply Object Oriented Programming Concepts to solve programming problems. Specifically, they will support the student's progress in the Laboratory.

Quizzes

Quizzes will be designed to test the student's understanding of the course and to assess various course outcomes.

Examinations

The exam shall contain problems designed to measure the ability and knowledge of the student to recognize, analyze, design, and apply Object Oriented Programming Concepts to solve programming problems.

Laboratory Work

The students will solve problems related to using and implementing OOP concepts in the Lab using appropriate language. The students will be assessed during each Lab session via Lab performance and Lab submissions in each Lab Session.

Textbook:

- 1 Python Crash Course: A Hands-On, Project-Based Introduction to Programming by Eric Matthes, 2nd Edition, ISBN-13:9781593279288 .
- 2 Introduction to Computation and Programming Using Python by John V. Guttag, 2nd Edition, ISBN-13 : 978-0262529624 .

Supporting Tools:

Lecture videos, notes, and other contents(Supplementary videos, slides) are available in BuX.

Course Policies:**Class Policy**

- Classroom and laboratory attendance are mandatory. You should come to the classroom before the instructor. Latecomers may/ may not be allowed to enter the classroom. Students, who are absent over 30% of the class time will not be allowed to enter the final examination
- You should turn off your cellular phone before entering the classroom. You should not leave the classroom to make or take cellular phone calls
- You should bring a notepad and/or a writing instrument to every class and take detailed notes.
- You should pay attention to the instructor and participate in class discussions.
- You should not do other work during class time.

Honor Code

Any form of cheating, plagiarism, and/or academic dishonesty will result in an "F" grade in the course.

Late Work and Examinations

Late assignments will not be accepted. Students who know that they are going to miss class should make arrangements in advance. Exams will be closed book. There will not be any make-up for quizzes and midterm exams except the cases of hospitalization or detention

Grading Policies:

Students' grades are assigned according to the grading scale of the Brac University Undergraduate Study and Examinations Regulations. In addition, the faculty are allowed to take into consideration the class average and standard deviation to reflect the actual class performance for student grade assignment. The grades at the university will be indicated in the following manner:

Marks	Grades
96>	A+ (4.0)
90-96	A (4.0)
85- <90	A- (3.7)
80- <85	B+ (3.3)
75- <80	B (3.0)
70- <75	B- (2.7)
65- <70	C+ (2.3)
60- <65	C (2.0)
57- <60	C- (1.7)
55- <57	D+ (1.3)
52- <55	D (1.0)
50- <52	D- (0.7)

<50	F (0.0)
P	Pass
I	Incomplete
W	Withdrawal
R	Retaken

Course Assessment Methods:

The following assessment methods are based on Theory Course only.

Section	Marks (%)
1. Participation in class	5 %
2. Quizzes/Class Tests	15%-20 %
3. Mid-Term Examination	20 % - 25 %
4. Lab & Project	20 % - 30 %
5. Final Examination	30 % - 40 %
Total	100 %

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